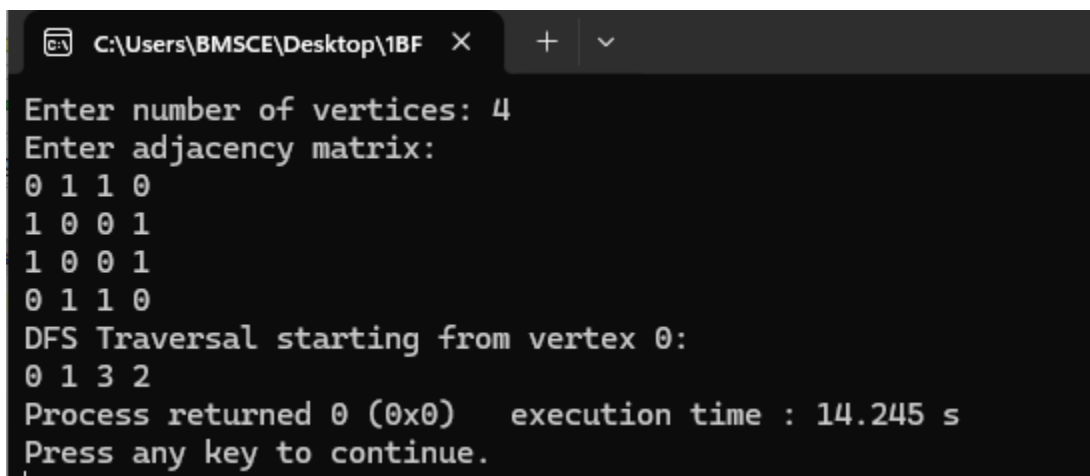


9B.DFS

```
#include <stdio.h>
#define MAX 10
int visited[MAX];
int adj[MAX][MAX];
int n;
void DFS(int v){
    visited[v]=1;
    printf("%d ",v);
    for(int i=0;i<n;i++){
        if(adj[v][i]==1 && !visited[i]){
            DFS(i);
        }
    }
}
int main(){
    printf("Enter number of vertices: ");
    scanf("%d",&n);
    printf("Enter adjacency matrix:\n");
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            scanf("%d",&adj[i][j]);
        }
    }
    for(int i=0;i<n;i++)
        visited[i]=0;
    printf("DFS Traversal starting from vertex 0:\n");
    DFS(0);
    return 0;
}
```

OUTPUT:



```
C:\Users\BMSCE\Desktop\1BF X + v
Enter number of vertices: 4
Enter adjacency matrix:
0 1 1 0
1 0 0 1
1 0 0 1
0 1 1 0
DFS Traversal starting from vertex 0:
0 1 3 2
Process returned 0 (0x0)   execution time : 14.245 s
Press any key to continue.
```